Who Gets Their Money Back? Predicting Outcomes and Mapping Themes in CFPB Consumer Complaints

Introduction and Overview

When someone files a complaint with the Consumer Financial Protection Bureau, the company's response gets recorded in a public database, including whether the complaint was closed with monetary relief. This project asks two questions. On the supervised side, can I predict whether a complaint will end in monetary relief from its narrative text and structured metadata, and which features actually carry that signal? On the unsupervised side, what themes live in the complaint narratives beyond the CFPB's official product and issue categories, and how do those themes shift over time and across companies? The two halves connect at the end, where I test whether the discovered theme features improve the outcome prediction model.

There are a few challenges I can see coming. Monetary relief is a small minority class, so everything about evaluation has to be imbalance aware. Credit reporting complaints have dominated recent volume, making up about 80 percent of complaints in fiscal year 2023, so I'll stratify my sampling and report results by product. Narratives are only published for consumers who opt in, which introduces selection bias, and the CFPB itself notes the database is not a statistical sample of consumer experiences, so my findings describe filed complaints rather than the whole marketplace. The label reflects the company's self reported closure category, not a neutral judgment of whether the complaint had merit. The narratives also contain XXXX redaction tokens and a lot of templated credit repair boilerplate that will need cleaning, and the product and issue options changed in 2017 and again in 2023, which complicates comparisons across years.

Closest Existing Work

The closest existing work is a 2026 paper titled From Complaint Narratives to Monetary Relief: A Hybrid Machine Learning Framework for CFPB Consumer Complaints, available at https://arxiv.org/abs/2606.22664. It predicts monetary relief by feeding LDA topic representations into machine learning classifiers as features, and it builds on a 2019 study by Bastani, Namavari, and Shaffer that applied LDA topic modeling to CFPB complaint narratives and tracked topic trends over time. My project differs in three ways. First, I treat feature representation as the main experimental variable, comparing structured only, TF-IDF, and transformer embedding representations head to head across several model families instead of committing to LDA features. Second, I add a temporal robustness angle by training on earlier years and testing on the post 2023 credit reporting surge, with calibration and drift checks. Third, my unsupervised half benchmarks classical topic models against modern embedding based clustering and looks at theme stability, human interpretability, and how relief rates differ by company within a theme. This is not a continuation of any previous Milestone or course project, and no supervised or unsupervised code is being reused from anywhere.

Part A: Supervised Learning

Dataset. The CFPB Consumer Complaint Database at https://www.consumerfinance.gov/data-research/consumer-complaints/. All the complaint data is published freely for anyone to use, analyze, and build on, with a single bulk CSV download at https://files.consumerfinance.gov/ccdb/complaints.csv.zip and a field reference at https://cfpb.github.io/api/ccdb/fields.html. It holds several million complaints going back to December 2011 and generally updates daily. Structured fields include product, sub product, issue, company, state, submission channel, dates, company response, and a timely response flag, plus free text narratives on a large opt in subset.

Approaches and feature representations. The target is binary, complaints closed with monetary relief versus other closed responses. I'll compare three feature representations. The first uses structured metadata alone, the second uses TF-IDF text features from the narrative, and the third uses pretrained sentence transformer embeddings. Each representation gets run through three model families, a logistic regression with L2 regularization as the interpretable baseline, gradient boosted trees with LightGBM, and a small neural network on the embeddings, with a majority class predictor as the floor. That grid covers linear, tree based, and neural approaches over both sparse and dense text encodings, which is exactly the comparison the project calls for. I'll handle imbalance with class weights and threshold tuning, and my splits will be temporal, training on earlier complaints and testing on recent ones so the setup matches how a model like this would actually get used.

External datasets and tools. I may join FDIC BankFind institution data to normalize company level results by institution size, which requires fuzzy company name matching as prep work. Tooling is pandas, scikit-learn, LightGBM, sentence-transformers, and SHAP.

Evaluation and visualization. PR-AUC is the primary metric given the imbalance, alongside ROC-AUC, F1 at a tuned threshold, and calibration curves, with everything sliced by product, company, and year. Planned visuals are precision recall curves per model and representation, calibration plots across time periods, SHAP summaries, and relief rate heatmaps by company and product.

Part B: Unsupervised Learning

Dataset. The same database, restricted to complaints with published narratives. Same links as above.

Structure questions. What recurring themes show up in the narratives beyond the official taxonomy? How does theme prevalence drift over time, for example pandemic era forbearance or the credit reporting surge? And do companies differ in their theme mix, with different relief rates attached to different themes?

Data manipulation. Filter to the narrative subset, then remove near duplicate templated complaints using MinHash or embedding similarity. Strip the redaction tokens, lemmatize for the

count based models, and compute sentence embeddings for the clustering track. I'll work from a stratified sample of roughly 200 thousand documents drawn by product and year to keep things tractable.

Approaches and representations. Classical topic models, LDA and NMF on TF-IDF, benchmarked against embedding based clustering using sentence embeddings with UMAP and HDBSCAN, plus k-means as a comparison point. Topic and cluster counts get chosen with standard coherence scores and density based validity measures.

External tools. gensim, BERTopic, umap-learn, hdbscan, and sentence-transformers.

Evaluation. Topic coherence for the topic models, silhouette and density based scores for the clusters, and stability measured with adjusted Rand index across bootstrap resamples. I'll also hand label a small validation sample to check that themes are human interpretable, measure alignment between discovered themes and the official product and issue taxonomy with normalized mutual information, and run the downstream test of whether theme features add lift to the Part A model.

Visualizations. Theme prevalence over time as a streamgraph, a UMAP scatter colored once by cluster and once by official product to show where they agree and disagree, top term charts per theme, and a company by theme relief rate heatmap.

Team Planning

Working solo, I own every piece: data acquisition and cleaning, deduplication, the supervised model grid and temporal splits, the topic modeling and clustering work, the evaluation framework and visualization suite, and all report writing and repo management. The single dataset shared across both parts and the heavy reuse of one preprocessing pipeline keep the scope realistic for one person.

Rough timeline. Weeks 1 and 2 cover the proposal, peer reviews, data download, cleaning, and EDA. Weeks 3 and 4 are the Part A representation and model grid, plus the first coach meeting and standup one. Weeks 5 and 6 are Part B topic models and clustering, plus standup two. Week 7 is integration, feeding theme features into the supervised model and running the disparity analysis, plus the second coach meeting. Week 8 is final report writing, figures, repo cleanup, and submission.